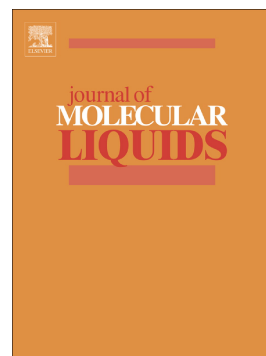


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A computational study of the ion gels formed by biodegradable aliphatic CBNAILs and BN nanostructures

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Abstract

A molecular dynamics (MD) study on the behavior of some ion gels consists of two biodegradable aliphatic cholinium-based naphthenic acid ionic liquids (CBNAILs) confined in the BN nanostructures has been reported in this work. The structural and dynamical properties of these ILs between two *h*-BN nanosheets (*h*-BNNSs) as a function of intersheet separation were analyzed. Also, the confinement effects of these CBNAILs inside the BN nanotubes (BNNTs) as a function of nanotube diameter along with the solvation effects of these ILs outside the BNNTs were studied. The results of radial distribution functions (RDFs), number densities, charge densities, the number of hydrogen bonds, coordination numbers, mass density contours and also diffusion coefficients showed that how the structure and dynamics of these ILs changes under confinement conditions. Generally, the strength of ion–BN nanostructure interactions decreases with increasing nanopore size for both anions and cations and the dynamics of confined ions are slower than that of the bulk fluid, i.e. the ions move faster into the larger nanopores (*h*-BNNSs or BNNTs) than the smaller ones. The results have been compared with the recent studies of the adsorption of these ILs on one *h*-BN sheet and also with the behavior of confined aromatic CBNAILs. The results showed the ability of the biodegradable aliphatic CBNAILs to interact with BN nanosystems for developing specific applications such as super capacitors, lubricants, gas adsorption, dispersion of nanostructures, etc.

Keywords: Biodegradable aliphatic cholinium-based naphthenic acid ionic liquids (CBNAILs); *h*-BN nanosheets (*h*-BNNSs); BN nanotubes (BNNTs); Ion gels; Structural properties; Dynamical properties.

1. Introduction

Ionic liquids (ILs) have been attracting more attention in recent years due to their interesting characteristics such as low vapor pressure, nonflammability, recyclability, high thermal and chemical stability. Moreover, the promising potential of this type of solvents can be tuned through a suitable combination of involved ions [1]. Based on these properties, ILs can be used in many fields [2-10]. ILs show unique behavior under confinement in cavities [11]. Most of the relevant studies on the ILs under confinement have been carried out in carbon nanosystems such as graphene [12-18], fullerene [18-20] and carbon nanotubes [17-18, 21-31]. Interest to the confined ILs is due to their technological applications such as heat-transfer fluids [32], supercapacitors [33-34], electricity generation [35], CO₂ capture [36], lubricant [37] and solvent for dispersion [23]. Thermal phase transition [38-41], thermal stability [42-43], vibrational properties [40, 44-46], fluorescence/luminescence spectra [40,47] and structural and dynamical properties [39,48] of ILs have been found to change in confinement. For example, the studies on the behavior of an IL confined inside a carbon nanotube have shown the development of crystal-like phases with high melting points [42,49-50]. Indeed, ionic arrangement for confined ILs lead to highly ordered structures with strongly dependent on temperature, nanotube diameter and the type of involved ions [51-52]. The changes in IL structuring in confinement may lead to some favorable properties such as increasing mobility for very viscous fluids [53-54].

Ion gels form a new class of hybrid materials, in which the properties of the IL are hybridized with those of another component such as carbon nanotubes, silica, polymer and inorganic fillers, etc [55]. The thermal stability and the negligible vapor pressure of the ILs makes ion-gels stable materials over time and allows for using them up to temperatures much higher than ambient. Actually, ILs act as structuring media during the formation of ion gels, their physicochemical properties and intrinsic organization influencing the building of the solid host network. It is expected that by controlling the properties and adjusting the interactions between IL and its host, smart ion gels with task specific can be envisioned.

Boron nitride (BN) nanostructures such as hexagonal BN nanosheets (*h*-BNNSs) and boron nitride nanotubes (BNNTs), comprised of equal number of alternating boron and nitrogen atoms, are different from carbon nanostructures in some aspects. *h*-BNNSs are intrinsically insulators, stable up to 1000 K and more resistant to oxidation than graphene [56]. Also, BNNTs exhibit high chemical stability, thermal stability (up to 800°C in air), excellent thermal conductivity, very high Young's modulus (up to 1.3 TPa), piezoelectricity, ultra-

violet light emission, etc. [57] These intriguing properties render BN nanostructures as ideal candidates for a variety of applications such as biology/medicine [58-59], cosmetics [60], mechanical or thermal reinforcements for polymer, ceramic and metallic composites, protective shields/capsules [61] and self-cleaning materials [62]. All these properties justify the great interest on BN nanostructures. It is very complicated to obtain a molecular-level picture of the behavior of ILs inside nanopores from experiments alone. A reliable computational model can assist in the interpretation of the experimental results and provide accurate predictions of the behavior of these systems. Molecular dynamics (MD) simulations and quantum mechanics calculations have been widely used to interpret the experimental observations and also to extend our insights into the features of different types of ILs at molecular level [63-69]. Also, these methods allow us to work with a porous material which is completely characterized. The study of the interactions of ILs with BN nanostructures is still scarce using both experimental [70-73] and theoretical approaches [74-77]. These studies are mainly limited to the classical ILs consisting of imidazolium or pyridinium cations and halide-containing anions for which remarkable toxicity and environmental problems have been reported in recent studies [78-79]. To our knowledge, the effects of confinement of ILs in BN nanostructures have not been investigated yet.

Choline based naphthenic acid ILs (CBNAILs) are a new family of ILs [80] with suitable properties such as null toxicity, high biodegradability and low cost which makes them as real green ILs. In our previous works [81-82], we carried out a computational study using both MD and density functional theory (DFT) methods to probe the properties of two CBNAILs with aliphatic rings in the pure state and also the adsorption properties of four CBNAILs on the *h*-BN surface. In this work, we present the first MD simulations to probe the behavior of some ion gels consist of two aliphatic CBNAILs, i.e. cholinium cyclopentane carboxylate ([CH][CPC]) and cholinium cyclohexane carboxylate ([CH][CHC]) confined between two *h*-BNNSs and also inside and outside of BNNTs. Fig. 1a shows the chemical structures of the cations and anions of these ILs. The effects of the size of the ring in the anion counterpart of the aliphatic ILs on the dynamical and structural properties of the studied CBNAILs under confinement have been investigated and the results have been also compared with aromatic CBNAILs.

2. Computational details

The LAMMPS molecular modeling package [83] was used for all the MD simulations reported in this work. ILs were described according to the parametrizations reported in the

CHARMM general force field [84,85]. Non-electrostatic model of the *h*-BNNSs and BNNTs were described according to the force field parametrization reported by Won and Aluru [86]. The force field parameters for the studied systems have been reported in Tables S1 and S2 of Supporting Information (SI). Armchair *h*-BN sheets and BNNTs were built using the nanobuilder plugin of VMD [87]. In the first part, boxes with 216 ion pairs were built for each IL and the force field parametrization for the studied CBNAILs was validated against experimental density data reported in the literature [80]. The deviations between simulated and experimental densities for [CH][CPC] and [CH][CHC] were 0.91% and 0.93%, respectively. Then, for each IL, the equilibrated IL boxes were confined between two fixed *h*-BNNSs with 5 Å distance from the upper and lower sheets (see Fig. 1b). The distance between the sheets was changed and the properties of the confined IL were analyzed at two different separations (i.e. 29.5 Å and 49 Å). The simulation details for the studied *h*-BNNSs have been given in Table 1. All of these simulations involving *h*-BNNSs were carried out in the NVT ensemble at 313.15 K using an equilibration of 10 ns followed by production run of 10 ns.

In the second part, the behavior of ions confined inside the nanotubes and the behavior of the ions in nanotubes' outer solvation shells in the armchair configuration were analyzed for the BNNT (10,10) and BNNT (15,15). Nanotubes were hydrogen terminated and their long axes were located along with the *z* axis. The number of boron, nitrogen and hydrogen atoms, and also diameter (*D*) and length (*L*) of BNNTs have been given in Table 2. A box with 512 ion pairs was built for each IL and then ions from a central cylindrical region of size suitable to fit the BNNT were removed and BNNT was placed in this region (see Fig. 1c). BNNTs were fixed along the simulations and the ions were allowed to move in and out of the nanotube channel. Therefore, once these systems composed of empty nanotubes surrounded by equilibrated CBNAILs were built, simulations in the NPT ensemble at 1 bar and temperatures in the range of 303 to 453 K were carried out (for 6 ns) to analyze the filling rates as a function of temperature and nanotube diameter. Afterward, the properties of filled nanotubes solvated by CBNAILs were analyzed through NPT simulations at 403 K and 1 bar using 10 ns simulations.

In all simulations, the temperature was controlled by a Nosé-Hoover thermostat [88-89] and the particle-particle-particle-mesh (PPPM) [90] was handled to carry out the long range electrostatic interactions with an accuracy of 1×10^{-4} . Also, the short-range forces were calculated by a cutoff of 13 Å. The cross-correlation LJ parameters for the unlike atoms were

computed using Lorentz–Berthelot combination rules [91]. The velocity Verlet algorithm [92] was used to solve the equations of motions with the integration time step of 1 fs.

3. Results and discussion

3.1. The confinement of aliphatic CBNAILs between two *h*-BNNSs

In this section, we have investigated the structures and properties of the aliphatic CBNAILs confined between two parallel *h*-BNNSs. Fig. 2 shows the number density profiles of the ions of [CH][CPC] and [CH][CHC] confined inside the slit-like *h*-BN nanopores with two different widths ($H = 29.5$ and 49 Å). In this figure, the center of mass of the ions was used to determine the *z*-coordinates. For choline cation, the N atom and for anions, the carbons of the ring connected directed to the COO^- group (atoms C_{14} and C_{16} of CPC and CHC anions in the structures shown in Table S1) were considered as the center of mass of the ions. In all the profiles, the dense layers of the ILs near both the top and the bottom walls of the nanopore were observed due to the strong interactions between the surface and CBNAILs, as reflected by the positions and heights of the first peaks. It must be mentioned that, because the non-electrostatic model was employed for BN sheet, the maxima of the anion peak and cation peak are not shifted and both cation and anion stick approximately to the walls. These results are in agreement with the results of Rajput et al. for $[\text{EMIM}^+][\text{TFMSI}^-]$ confined in a uncharged slit nanopore [14]. The height of the peaks in the density profiles decreases when the distance between the two *h*-BNNSs increases. In addition, with increasing the distance, the number of layers of the cation and anion increases. At the center, due to the bulk behavior, sharp peaks were not seen. These observations are similar to some other simulation studies [13-14, 18]. As can be seen from Fig. 2a, for both pore sizes, the high-density layers appear for both ions of [CH][CPC] at almost the same distance of the two *h*-BNNSs. The distances between the first peak of the ions and the top and bottom walls of smaller pore are ~ 3.75 Å and 4.25 Å, respectively and these values for the larger pore are ~ 3.5 Å and 4.5 Å, respectively. The density profiles for [CH][CHC] (Fig. 2b) exhibit that the [CH] cations are closer to the surface than those of the anions for both pore sizes. The locations of peaks in this work are almost the same as those of observed in our previous work [82] and also with the observations of Aparicio and Atilhan [13]. In our previous work [82], for the adsorption of aliphatic CBNAILs on one *h*-BN surface, the density profiles showed that the maximum number density of the first adsorbed layer for all of the ions was located at ~ 4 Å from *h*-BN surface. Aparicio and Atilhan [13] obtained the first peak for the ions of [MP][LA] on

graphene surface and confined between two graphene layers at about 3.5 Å from the surface. A comparison between the cation and anion profiles in both pore sizes shows that a second-high density layer for [CH] cation in the aliphatic CBNAILs appears at ~10 Å of walls of larger nanopore, whereas this second layer is poorly observable in the smaller pore size. The values of intensities of the first peak for the ions of [CH][CPC] and [CH][CHC] in large separations between the surfaces ($H = 49$ Å) are very similar to those of aliphatic CBNAILs on one *h*-BN surface [82]. Also, it can be inferred from Figs. 2a,b that for larger pore size ($H = 49$ Å), increasing the number of carbon atoms in the ring of [CHC]⁻ with respect to [CPC]⁻ causes an increment in the thickness of the first peak region. It can be also concluded from these figures that the number of cations at the first adsorbed layer decreases with increasing the size of [CHC]⁻ with respect to [CPC]⁻ probably due to the more packing density in [CH][CPC]. Generally, the density profiles of aliphatic CBNAILs confined inside larger nanopore ($H = 49$ Å) are very similar to those reported in our previous work for adsorption on one *h*-BN surface [82]. In the other words, the confinement effects can be seen more obvious in the separation distance of 29.5 Å. The structuring in the vicinity of *h*-BNNSs upon confinement for aliphatic CBNAILs is confirmed by the charge density profiles reported in Figs. 3a,b. Fluctuations in the ion charge concentrations confirm the layered structure of the ILs. Similar to number density profiles, because of non-electrostatic model, both cation and anion stick approximately to the walls and opposite charge layers are not be seen. As can be seen from these figures, the maxima and minima of these peaks decrease with increasing H from 29.5 to 49 Å, especially for those corresponding to the first adsorbed layer.

The distribution of the studied aliphatic CBNAILs confined between two *h*-BNNSs in different interwall distances may be analyzed using the density contour plots depicted in Fig.4 (after 10 ns of simulation time). In this figure, the green and red color codes represent the low and high mass density regions, respectively. The maximum density of the adsorbed layers for all of the ions is located near *h*-BNNSs and the central region shows lower density. Because of 5 Å distance between ILs and BN sheets (as mentioned in the computational details section) and the interactions between ILs and sheets, the number of ILs in the middle of the box decreases. Different patterns are observed for aliphatic CBNAILs confined inside the nanopores with two different widths. According to the Figs. 4a,b, at $H = 49$ Å, two *h*-BNNSs are covered by several adsorbed layers of ILs for both of the studied aliphatic CBNAILs. But Figs. 4c,d show that at $H = 29.5$ Å, two *h*-BNNSs are covered by approximately one adsorbed layer of ILs. It seems that the increase of ion numbers in the

solid-like (high-density) region near the *h*-BNNSs at $H = 29.5 \text{ \AA}$ leads to a reduction in the number of layers. This result is in agreement with the number density profiles in Fig. 2.

Radial distribution functions (RDFs) can give beneficial information about the structure of confined aliphatic CBNAILs. Fig. 5 shows the scaled RDF plots (corrected for excluded volume effects [93-96]) of cation–anion for the different layers of [CH][CPC] confined between two *h*-BNNSs with the width $H=49 \text{ \AA}$ and also similar RDF computed for the simulated bulk IL at the same temperature (313.15 K). It must be mentioned that the excluded volume effects do not change the location of the peaks of RDF and mainly effects on the height of the peaks and the value of RDF at large distances. In fact, these effects do not produce additional “artificial” peaks in RDF [16]. From the density profile (see inside Fig. 5), the confined ions were divided into three different layers: the first layer (closest to the pore walls), the second layer, and the middle layer. The first and second layers consist of the molecules in which their *z* positions are in the range of 2.8 to 6.8 $\text{\AA}$ and 6.8 to 12 $\text{\AA}$ from the bottom wall, respectively. The middle layer, is intermediate between two confining surfaces from 12 $\text{\AA}$ to 38 $\text{\AA}$. Fig. 5 shows that in all of the layers and also in the bulk, the peaks of the cation–anion RDF appear almost at the same distance (5.15 $\text{\AA}$). The results indicate that confinement leads to increase in the height of the RDF peaks as compared with those of the bulk IL. Generally, the height of the peaks in cation–anion RDFs of [CH][CPC] obeys the following order: first layer > second layer > middle layer > bulk. The intensity of the peaks decreases with increasing distance from the surface, due to the increase of the available space for the movement of ions and decrease of the compactness of the ILs. Thus, the results showed that the cation–anion interactions near the *h*-BN sheet are stronger because of the aggregation of ILs near the sheet. Similarly, Atilhan and Aparicio [97] showed the center-of-mass RDFs of [EMIM][DCA] on top of graphene surface from MD simulations at 298 K. Their results showed that the basic structure of peaks (positions of the maxima, relative widths) is maintained in the four regions considered, even in *zI* region, which is in direct contact with the graphene surface, and thus, confirming that ion pairs are not disrupted upon adsorption on the graphene surface. Analogous results are also obtained for the confined [CH][CHC] IL.

The individual atomic RDFs between B and N atoms of the *h*-BN surface and several atoms of the studied aliphatic CBNAILs have been studied and discussed in our previous work [82] and also compared the behavior of these CBNAILs with some aromatic ones. In our previous work, we concluded that in spite of CBNAILs with aromatic rings in which anions tend to be

arranged in parallel orientation to the surface, the orientations of anions in the aliphatic CBNAILs are completely non-parallel. The crucial interactions between the aromatic anions and *h*-BN sheet arise from the π - π interactions between the aromatic ring and the surface, while there aren't such interactions for the aliphatic anions with the BN sheet. Also, this study showed that the anions doesn't stick to the walls by means of the COO^- groups (see Fig. 6 of Ref. [82]). In this work, for better understanding of the structural properties of the studied CBNAILs under confinement, the profiles of scaled RDFs between B atoms of the *h*-BNNSs and the center of mass (COM) of the ions of ILs have been illustrated in Fig. 6a (for anions) and in Fig. 6b (for cations) for the nearest layer to the confining surfaces at different distances between two sheets at 313.15 K. The RDFs between N atoms of the *h*-BNNSs and the COM of the ions of ILs were similar to those shown in Figs. 6a,b. Overall, the strength of ion-surface interactions decreases with increasing nanopore size for both anions and cations. This observation clearly indicates the more aggregation of the ions at smaller distances ($H=29.5$ Å) and confirms the effect of this compactness on the intensity of the peaks. This result is in agreement with the behavior of imidazolium-based ionic liquids under confinement [12]. As can be seen, the height and the location of maxima in anions and cations are almost the same in both nanopores. The comparison between Figs. 6a and 6b (for $H = 29.5$ Å) shows that the first peak of the cation of [CH][CHC] appears at smaller distances than that of the [CHC] anion and also their intensity is stronger, which shows that the interaction of cation in [CH][CHC] with surface is stronger than the corresponding interaction of anion with surface. These results are in agreement with the results of their number density profiles in Fig. 2.

Fig. S1 shows the scaled RDFs of cation-anion for the first layer of [CH][CPC] confined between two *h*-BNNSs with the variations of the interwall distances from 29.5 to 49 Å at 313.15 K. In both of the confined systems, the first peak of the cation-anion RDFs appears almost at the same distance. According to this figure, the cation-anion interactions in the first layer of the confined system with $H=29.5$ Å decreases significantly with respect to the confined system with $H=49$ Å. In fact, the stronger interactions of IL/BNNSs in the system with $H=29.5$ Å (see Fig. 6) cause the weakness of the cation-anion interactions in this system.

To find out how the hydrogen bonds among the ions of [CH][CPC] can get affected under confinement, the average number of the hydrogen bonds between the cations and anions of [CH][CPC] in bulk and confined systems as a whole, in the first layer of confined systems

and in different layers of confined system with $H = 49 \text{ \AA}$ have been given in Fig. 7. In Fig. 7a, the number of hydrogen bonds between the confined ILs as a function of the distance between two sheets and bulk (in the absence of surface) were calculated. The hydrogen bondings of [CH][CPC] were calculated considering H atom of hydroxyl group in choline as donor and O atoms in [CPC] as acceptor. The results show that the number of hydrogen bonds among [CH][CPC] in the confined systems is less than that of the bulk case due to the disruption in the network of hydrogen bonds among ions in the presence of confining BN surfaces. Also, the number of hydrogen bonds between anions and cations increases with increasing the distance between the confining BN sheets. In fact, the stronger interactions between ILs and BNNSs in the confined system with $H=29.5 \text{ \AA}$ weaken the average cation–anion interactions which in turn decrease the number of hydrogen bonds in this system.

We have also calculated the average number of the hydrogen bonds between the confined cations and anions of the first layer of [CH][CPC] at different pore sizes ($H = 29.5 \text{ \AA}$, $49 \text{ \AA}$). According to Fig. 7b, the average number of hydrogen bonds decreases with decreasing the pore size. Again, it seems that the stronger interactions between ILs and BNNSs in the first layer of the system with $H=29.5 \text{ \AA}$ cause the weakness of the cation–anion interactions in this system. Moreover, the number of hydrogen bonds of different layers of [CH][CPC] was computed. The results show that the average number of hydrogen bonds decreases from the first layer in direct contact with *h*-BNNSs to the middle layer (See Fig. 7c). It indicates that the compactness of [CH][CPC] in the first layer near the *h*-BNNS leads to the increase in the average number of hydrogen bondings. This is in agreement with the results obtained by Atilhan et al. for [EMIM][DCA] [97].

Studying the translational dynamics is important for fundamental understanding of conductivity, mass transfer, interfacial behavior and rheology of liquids. The translational dynamics of the confined ILs can be studied by measuring their mean square displacement (MSD). At $T=313.15 \text{ K}$, the dynamics of the studied confined systems is very slow and after 10 ns simulation, the MSD plots didn't reach to the diffusive regime. The β values in diffusive regime should reach a value equal to one, in which the MSD of the ion increases linearly with time. The β parameter can be obtained as

$$\beta(t) = \frac{d \log(MSD)}{d \log(t)}$$

(1)

However, at this low temperature and in this time period, the values of the β parameter are far from 1. So, we in Figs. 8 and S2 just displayed the graphs of MSDs to compare the dynamics of ions in different layers qualitatively.

Fig. 8 shows the parallel (-x and -y) and also the z- components of the MSD of the different layers of [CH][CPC] confined inside the BN nanopores with width 49 Å at 313.15 K. As expected, the dynamics of ions in the xy plane are faster than those in the z direction. This may be due to the geometric distance for movement in direction z which is less than that for movement along other axes. Also, the ions in the first layer have slower dynamics than those in the second layer. Therefore, the dynamics of the confined ions depends strongly to their distance from the walls. In the first layer, the strong interactions between the ions and surface, cause to slow the dynamics. In all regions of the pore, cations move faster than anions, which may be due to the lower relative mass of $[\text{CH}]^+$. Fig. S2 shows the parallel (xy) and normal (z) components of the MSDs of the cations and anions of the studied aliphatic CBNAILs for the first layer between two confining surfaces ($H = 29.5 \text{ Å}, 49 \text{ Å}$) at 313.15 K. The results indicate that the mobility of the confined ions increases with increasing pore size from 29.5 to 49 Å, because the strength of ion-surface interactions decreases with increasing nanopore size. As Fig. S2 shows, the effects of confinement on the mobility of molecules along the z axis are more observable. The lower slope in z-direction may be due to the decreasing in mobility of ions as a result of the interaction with surfaces. The slowest dynamics are observed for the ions of [CH][CHC] in the z-direction and the dynamics of [CH][CHC] is slower than [CH][CPC] in both z- and xy- directions. Indeed, with an increase in the number of carbons in the anion ring, the total strength of the vdW interactions enhances while the dispersion of the charge centers causes weakening the electrostatic attractions between the anion and cation. The balance between these two opposing effects determines the dynamics of ions. It seems that the strength of vdW interactions dominate and the MSDs of both the cation and anion in [CH][CHC] decreases.

3.2. Aliphatic CBNAILs inside and outside BNNTs

In this section, the behavior of aliphatic CBNAILs with respect to BNNTs (10,10) and (15,15) was studied. Fig. 9 shows the filling rates of empty nanotubes surrounded by the ions of [CH][CPC] and [CH][CHC] in the range of 303 to 453 K for both selected BNNTs. Also, the number density values of cations and anions within BNNTs were reported in Table 3. Comparing both BNNTs, it can be found that the larger diameter nanotubes can

accommodate more ions and the entering speeds of ions increases with increasing temperature. Indeed, the dynamics of ions penetrating inside the nanotube is largely depend on the temperature. This may be due to the decreasing of viscosity of these ILs with increasing temperature. For example, the viscosity of [CH][CPC] decreases from 306.5 mPa.s at 303.15 K to 49.9 mPa.s at 353.15 K [80]. Fig. 9 shows that the saturation time of nanotubes decreases with increasing temperature for both studied ILs. Also, the maximum number of cations and anions within the nanotubes has been reached at times lower than 1ns. As Fig. 9 shows, there isn't any significant difference between the filling rates of cations and anions for both of the studied ILs at each temperature (for example see Figs. 9a and 9b). This is in spite of the results obtained by Dong et al. [98] in which they reported that cations are always faster than the anions in the whole filling process for [BMIM][PF₆] confined inside single-walled carbon nanotubes (SWNTs) (9,9) and (10,10). According to Table 3, in BNNT (15,15), the number density of cations and anions of [CH][CPC] is more than that of the [CH][CHC] which may be attributed to the lower viscosity and also smaller size of [CH][CPC]. Also, to compare the behavior of aliphatic CBNAILs versus the aromatic ones, i.e. choline salicylate ([CH][SA]) and choline benzoate ([CH][BE]), we calculated the values of cation and anion number densities for these aromatic ILs inside nanotubes in the range of 303-453 K (see Table 3). As this table shows, at lower temperatures the number densities of aromatic CBNAILs are much smaller than those of the aliphatic ILs. This may be due to the slow dynamics of aromatic ILs which have high viscosities. At high temperatures (403 K and 453 K) in which the dynamics is fast, the number densities of aromatic ILs are even more than the aliphatic ones. This opposite behavior at high temperatures may be due to the planar structure of anion in aromatic ILs which causes the increase in the number of ions within the nanotubes.

The confinement of aliphatic CBNAILs leads to a reduction in number density compared to the bulk liquid phase. The calculated number density for pure [CH][CPC] and [CH][CHC], calculated in the absence of BNNTs, is 2.64 nm^{-3} and 2.45 nm^{-3} at 353 K, respectively. Although ionic densities within the BNNTs increase with n , these densities are lower than that of the bulk ILs. For example, at 353 K, the relative number densities of [CH][CPC] under confinement with respect to the bulk system are 53% and 58% for BNNT (10,10) and BNNT (15,15), respectively. For [CH][BE], as an aromatic IL, the ion concentrations within the BNNT are 33% and 60% of the bulk IL values for BNNTs (10,10) and (15,15), respectively which are in agreement with the results of Aparicio et al. [21]

The structuring of ions within the BNNTs may be analyzed from the number density profiles along the nanotube z -axis. Figs. 10a-d show the distribution of cations and anions of aliphatic CBNAILs along the axial direction inside and outside of the BNNTs (10,10) and (15,15) at 403 K. Also, snapshots of ions confined inside the BNNTs have been represented at the top of these figures to give a clear image of such distributions. For [CH][CPC], the z distributions in the interior of BNNTs vary with the nanotube diameter. In the case of $n = 10$, Fig. 10a, the axial distributions show seven peaks separated by around 5 Å for both ions. The z distributions for [CH] and [CPC] suggest that the cation and anion pairs are situated at the same locations. There are about 6-7 such pairs inside the BNNTs (10,10) (see Fig. 10). Snapshot at the top of the Fig. 10a confirms that the cations and anions form zigzag arrangement inside the nanotube (10,10). The structuring of [CH][CPC] within the BNNT (15,15) is different. In this case, the pattern of distribution of ions show periodic oscillations. This trend can be understood in terms of the increase in available space which causes an enhancement in entropy.

In the case of [CH][CHC] (see Figs. 10b,d), the zigzag arrangement is not observed within BNNT(10,10). Indeed, the arrangement of ions is characterized by alternating anion and cation distributions. Also, the pattern of axial distribution inside the BNNT (15,15) is almost the same as that of the BNNT (10,10) but with larger concentrations. Similarly, Aparicio et al. [21] reported that the axial distribution of [CH][SA] and [CH][BE] are almost the same inside the SWNTs (10,10) and (15,15). The formation of ordered structures inside nanotubes has been confirmed by some other researchers [99-100]. For example, Shim and Kim [101] showed that for [EMIM][BF₄] confined inside SWNT (10,10) and SWNT (15,15), anions and cations developed as chiral and staggered pentagonal distributions, respectively. Also, Herrera et al. [18] reported that ions of [EMIM][GLY] have a zigzag arrangement in SWNT (10,10) and SWNT (15,15). As can be seen from Figs. 11a,b, charge distribution along the axial direction within the nanotubes shows that the total charge (anion + cation) is almost null within the nanotubes for both BNNT (10,10) and BNNT (15,15) which is in agreement with the results of Table 3 for number densities of ions of each CBNAILs. Also, oscillating behavior inside the nanotubes is in agreement with the ionic distribution reported in Fig. 10.

The density contour plots of axial and radial distribution depicted in Figs. 12a,b show the solvating of BNNT(15,15) with anions and cations and also the confinement of ions within the nanotube at 403 K. As can be seen, the presence of the BNNTs leads to remarkable changes in the arrangement of aliphatic CBNAILs confined within the nanotube channels and also for the ILs solvating the nanotubes. Results reported for anions and cations radial

distribution of [CH][CPC] (see two bottom plots of Fig. 12a) and [CH][CHC] (see two bottom plots of Fig. 12b) show that the distribution of cations and anions outside the BNNT walls is characterized by two well-defined spherical shells for both ILs, although the second shell is more obvious for [CH][CHC]. The formation of two layers around the nanotubes is similar to the distribution of [CH] and [BE] outside the SWNT at 353 K [21] and also with the distribution of the studied aliphatic CBNAILs near the sheets in nanopore with $H=49$ Å discussed in previous section (see Fig. 2). Also, these figures show one well-defined anion internal solvation shell for both [CPC] and [CHC] anions, with an empty space in the center of nanotubes. On the other hand, the distribution of cations inside the nanotubes for both ILs shows two well-defined layers. These observations are in contrast with the aromatic CBNAILs inside the SWNTs in which the cations have one well-defined solvation shell and anions have two-shells structure inside the nanotube [21]. The formation of several internal solvation shells has been also reported previously for imidazolium-based ionic liquids [7, 101-103].

The coordination number plots of aliphatic CBNAILs confined within the BNNTs with different diameters have been presented in Fig. 13. As this figure shows, for both ILs, with increasing in the diameter of nanotube from $n=10$ to $n=15$, coordination number increases. Indeed, the ions inside the nanotube (15,15) are less affected by wall, therefore, there are more cations around each anion. Also, the shape of the coordination curves for ILs inside the BNNT (15,15) is more similar to the bulk compared with the ILs within the BNNT (10,10). These results are in agreement with the aromatic CBNAILs confined inside the SWNTs (10,10) and (15,15) at 353 K [21].

Fig. 14 shows the RDFs of ions confined within the BNNTs as a function of distance from the BNNT wall at 403K. These figures confirm the formation of one single internal shell inside the BNNT (10,10) for both anions and cations. In the case of BNNT (10,10), the COM RDFs of [CHC] and [CPC] anions are placed at 4.60 and 4.79 Å from the nanotube walls (see Fig. 14a), respectively, whereas the COM RDFs of [CH] cations in both ILs are placed at 4.97 Å from the nanotube walls (see Fig. 14b). Thus in both ILs, for BNNT (10,10), anions are closer to the nanotube internal wall. As Fig. 14b shows, there are two obvious peaks at about 5 and 10 Å from the nanotube walls correspond to the two internal solvation shells depicted in Fig. 12. In the case of anions inside the BNNT (15,15), there is just one broad peak around 6 Å corresponds to one internal solvation shell which is observable in Fig. 12. Aparicio et al. [21] reported two internal shells at about 4.2 and 10.5 Å for [BE] anion inside

the SWNT (15,15) at 353K. This effect may be due to the planar structure and the parallel arrangement of aromatic anions near the walls.

Finally, the diffusion coefficients (D) were calculated using the Einstein's relationship from the MSD plots. The corresponding Einstein relation is

$$D = \frac{1}{6} \lim_{t \rightarrow \infty} \frac{d \text{MSD}}{dt} \quad (2)$$

In Table 4 we have presented the results of the diffusion coefficients and β values of the aliphatic CBNAILs confined inside the BNNTs (10,10) and (15,15) in the axial direction at 403 K. The reported results indicate that the diffusion rates for confined ions are slower than that of the bulk fluid and the ions move faster into the larger BNNT than the smaller one. The similar results reported by Singh et al. [103], Akbarzadeh et al. [49] and Aparicio et al. [13, 21] for IL molecules inside the carbon nanotubes. However, Herrera et al. [18] and Chaban and Prezhdoo [53] reported that the mobility of ions of [EMIM][Cl] and [EMIM][GLY] inside the carbon nanotubes is increased under confinement, i.e. the ions move faster into the smaller BNNT than the larger one. As can be seen, the diffusion coefficients of the cations are greater than that of the anions in all of the studied systems, which is in agreement with the behavior of ions in bulk fluid and also the results of ILs confined between two h -BNNSs in previous section. It seems that the lower relative mass of [CH] cation can justify the larger diffusion rates of cations. In contrast with the aliphatic CBNAILs, Aparicio et al. [21] reported that the aromatic anions ([BE] and [SA]) move faster than choline cation inside SWNT. This effect probably occurs because of the planar structure of aromatic anions which allows more mobility with respect to nonplanar [CH] cations. Also, the higher mobility of [CPC] anion compared to that of the [CHC] anion inside the nanotubes arises from the smaller size of the [CPC] anion.

4. Conclusion

The behavior of some ion gels consist of two aliphatic CBNAILs with respect to the BN nanosystems were studied in this work using MD simulations. In the first part, regarding the confinement of the studied CBNAILs between two h -BNNSs, the effect of sheet separation was investigated. In the number and charge density profiles, the dense layers of the ILs were observed near both the top and the bottom walls of the nanopore due to the strong interactions between the surface and ILs. Generally, the confinement effects can be seen more obvious in

the separation distance of 29.5 Å and the density profiles of aliphatic CBNAILs confined inside larger nanopore ($H = 49$ Å) are very similar to those reported in our previous work for adsorption on one *h*-BN surface.

The mass density contours showed that the *h*-BNNSs with the width $H=29.5$ and 49 Å are covered by one and two adsorbed layers of ILs, respectively. Indeed, the increase of ion numbers in the solid-like region near the *h*-BNNSs at $H = 29.5$ Å leads to a reduction in the number of layers. In the anion-cation RDFs, the intensity of the peaks decreases with increasing distance from the surface, due to the increase in the available space for the movement of ions and decrease in the compactness of the ions which is in agreement with the results of the average number of hydrogen bonds in different layers. Also, the results indicated that the mobility of the confined ions increases with increasing pore size from 29.5 to 49 Å, because the strength of ion-surface interactions decreases with increasing nanopore size.

In the second part, the confinement effects of these CBNAILs inside the BNNTs as a function of nanotube diameter were studied. The confinement of aliphatic CBNAILs leads to a density reduction compared to the bulk liquid phase and the larger diameter nanotubes can accommodate more ions. Also the entering speeds of ions increases with increasing temperature. For [CH][CPC], the axial distribution of ions in the interior of BNNTs change from zigzag arrangement to alternating with varying BNNTs from $n=10$ to $n=15$. For [CH][CHC], the pattern of axial distribution of ions inside the BNNT (15,15) is almost the same as that of the BNNT (10,10) but with larger concentrations. The distribution of cations and anions outside the BNNT walls is characterized by two well-defined spherical shells for both ILs. The RDF plots confirm the formation of a single internal shell inside BNNT (10,10) for both of the anions and cations and two internal solvation shells for cations inside BNNT (15,15). The results showed that the diffusion rates for confined ions are slower than that of the bulk fluid and the ions move faster into the larger nanotube than the smaller one.

Supporting Information

The supporting information file contains Table S1, force field parameters for [CH]⁺, [CPC]⁻, [CHC]⁻ ions and BN nanostructures; Table S2, The CHELPG atomic charges used for [CH]⁺, [CPC]⁻ and [CHC]⁻ ions; Fig. S1, The COM RDFs of anion-cation for [CH][CPC] confined between two *h*-BNNSs with different intersheet distances ($H = 29.5$ Å, 49 Å) at 313.15 K for the first layer; Fig. S2, Parallel (*xy*) components of the MSD for (a) the cations and (b) the

anions and z components of the MSD for (c) the cations and (d) the anions of the studied aliphatic CBNAILs in the first layer near the confining surfaces ($H = 29.5 \text{ \AA}$, $49 \text{ \AA}$) at 313.15 K.

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Figure Captions

Fig. 1. (a) Molecular structures of the ions of the studied CBNAILs. Atom color code: carbon (light blue), nitrogen (blue), oxygen (red) and hydrogen (light gray), (b) Initial configuration used in the MD simulation of [CH][CPC] confined between two *h*-BNNSs of width $H = 49 \text{ \AA}$ and (c) Initial configuration used in MD simulation of BNNT(15,15)-[CH][CPC] system. Analogous systems were built for [CH][CHC].

Fig. 2. Atomic number density profiles along the *z*-axis (perpendicular to the *h*-BNNSs) for the ions of (a) [CH][CPC] and (b) [CH][CHC] confined between two *h*-BNNSs with different distances. Results for CBNAILs confined between two *h*-BNNSs with width $H = 29.5 \text{ \AA}$ are shifted upward for the sake of visibility. The dashed lines show the position of the *h*-BNNSs.

Fig. 3. Charge density profile along the *z*-axis (perpendicular to the *h*-BNNSs) for the ions of (a) [CH][CPC] and (b) [CH][CHC] confined between two *h*-BNNSs with different intersheet distances. Results for CBNAILs confined between two *h*-BNNSs with width $H = 29.5 \text{ \AA}$ are shifted upward for the sake of visibility.

Fig. 4. Mass density contour maps for aliphatic CBNAILs confined between two *h*-BNNSs with width (a,b) $H = 49 \text{ \AA}$ and (c,d) $H = 29.5 \text{ \AA}$. The green and red color codes represent the low and high mass density regions, respectively.

Fig. 5. The COM RDFs of anion-cation for different layers of the [CH][CPC] confined between two *h*-BNNSs with the width $H=49 \text{ \AA}$ at 313.15 K. The same RDF for the bulk [CH][CPC] are also shown.

Fig. 6. The RDF plots of (a) the anions... *h*-BNNSs and (b) the cations... *h*-BNNSs in different intersheet distances at 313.15 K for the first layer.

Fig. 7. The average number of the hydrogen bonds between the cations and anions of [CH][CPC] at 313.15 K, (a) bulk and confined systems as a whole, (b) the first layer of confined systems, and (c) different layers of confined system with $H = 49 \text{ \AA}$.

Fig. 8. The MSD plots of ions for the different layers of the [CH][CPC] confined between two *h*-BNNSs with the width $H=49 \text{ \AA}$ at 313.15 K, (a) in the parallel direction and (b) in the perpendicular direction to the confining surfaces.

Fig. 9. Number of confined ions, N_C , inside BNNTs with different diameters as a function of the simulation time, t , in different temperatures at 1 bar.

Fig. 10. Number density profiles for ions along the axial direction, z , for BNNT (10,10) and BNNT (15,15) at 403K and 1bar. The dashed vertical lines show the position of the BNNT terminal hydrogen atoms. Also, snapshots of ions confined in the BNNTs along the axial direction have been presented upon each figure. (Only the center of mass of each ion are plotted to improve visibility). Color code: (red) $[\text{CH}]^+$, (orange) $[\text{CHC}]^-$ and (green) $[\text{CPC}]^-$.

Fig. 11. Charge density profiles of the ions along the axial direction of (a) BNNTs (10,10) and (b) BNNT (15,15). Vertical dashed lines show the BNNT walls.

Fig. 12. Mass density contour maps for the ions of (a) [CH][CPC] and (b)[CH][CHC] inside and outside of the BNNT (15,15) along the axial and radial directions. The green and red color codes represent the low and high mass density regions, respectively.

Fig. 13. The number of cations surrounding anions, N , for anions and cations of (a)[CH][CPC] and (b) [CH][CHC] inside the BNNTs with different diameters and also in bulk at 403 K and 1bar.

Fig. 14. The RDF plots for the center of mass of (a) anions and (b) cations confined inside the BNNTs with different diameters along the short axis of the BNNTs as a function of distance from the BNNTs wall at 403 K and 1bar.

Fig. 1.

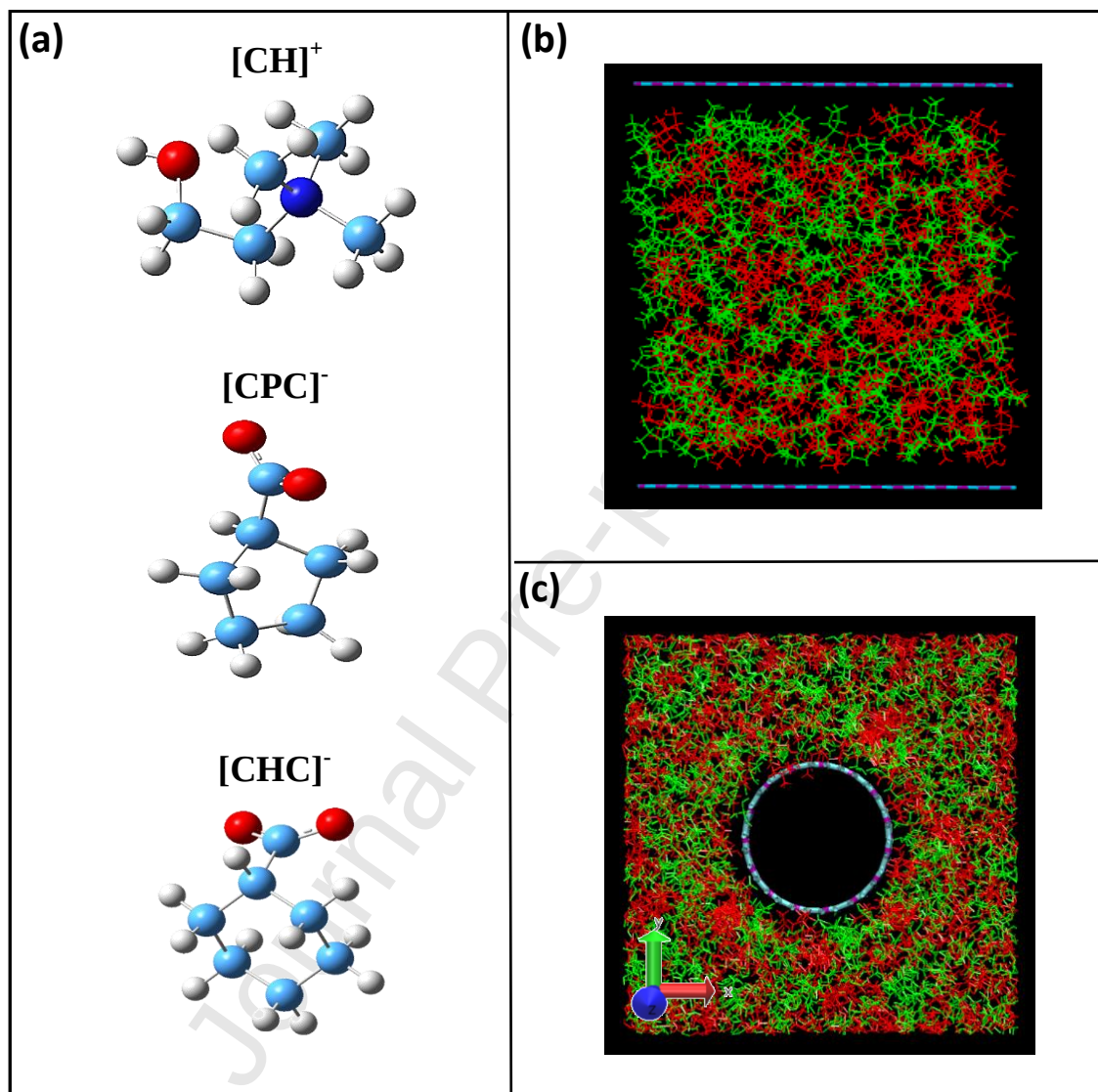


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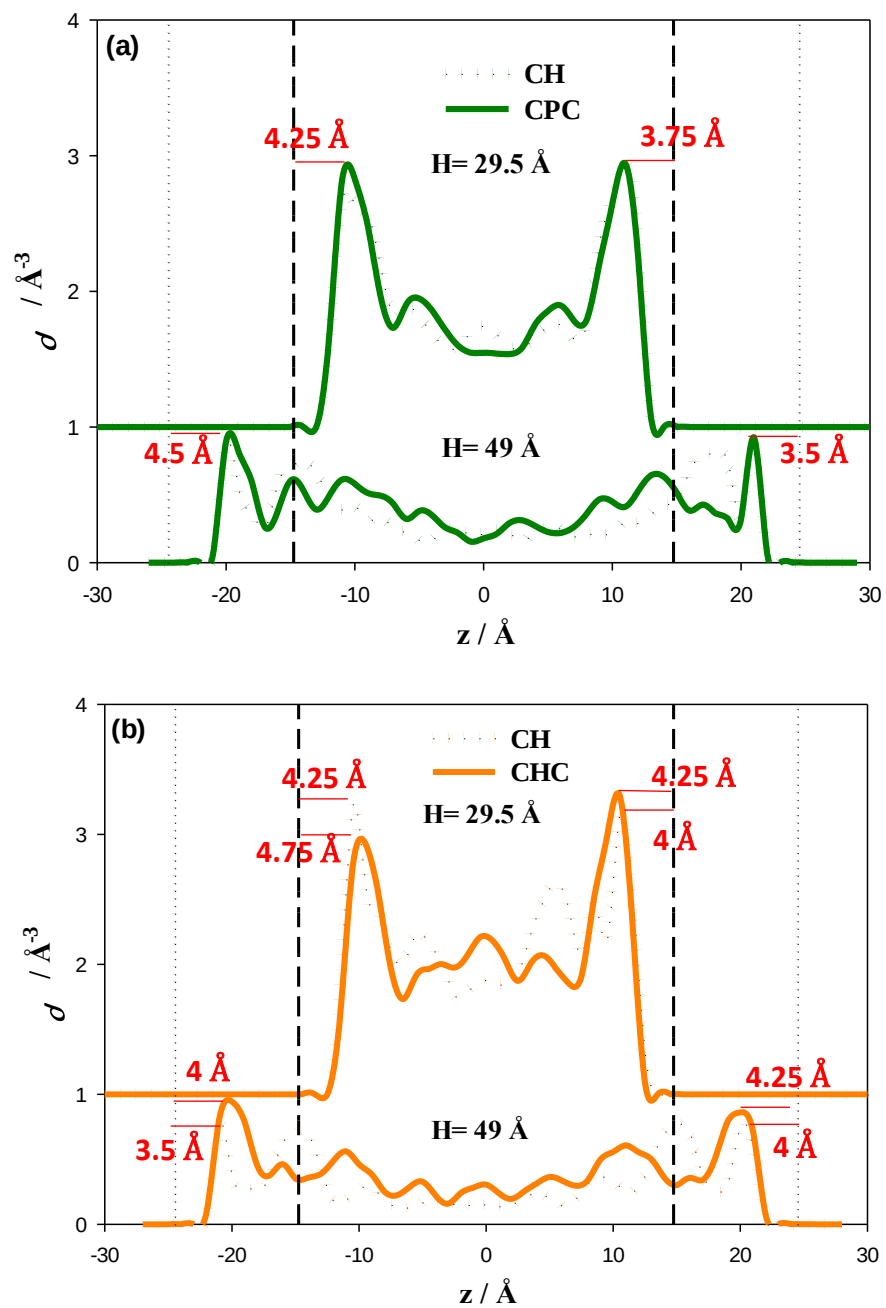


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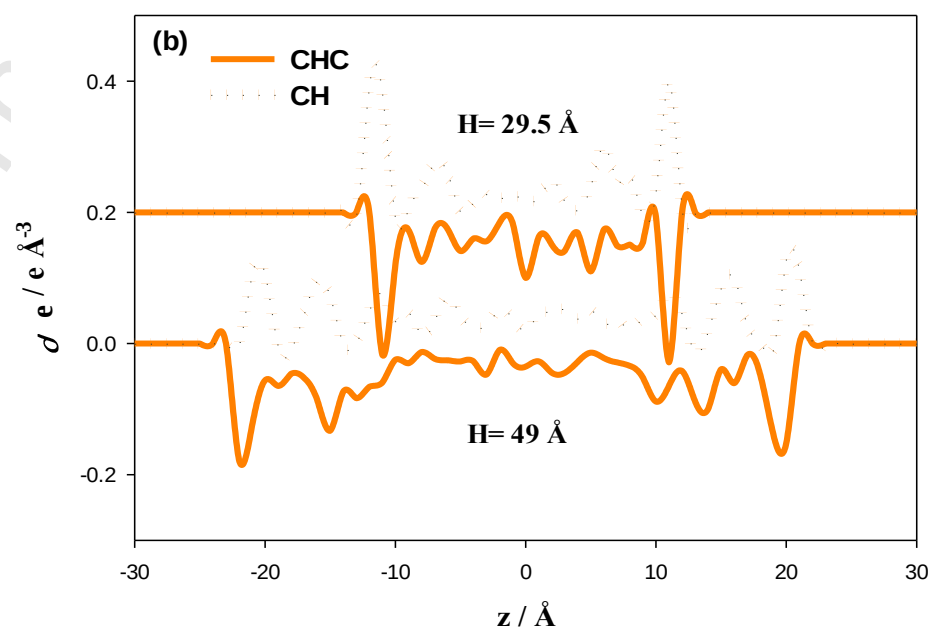
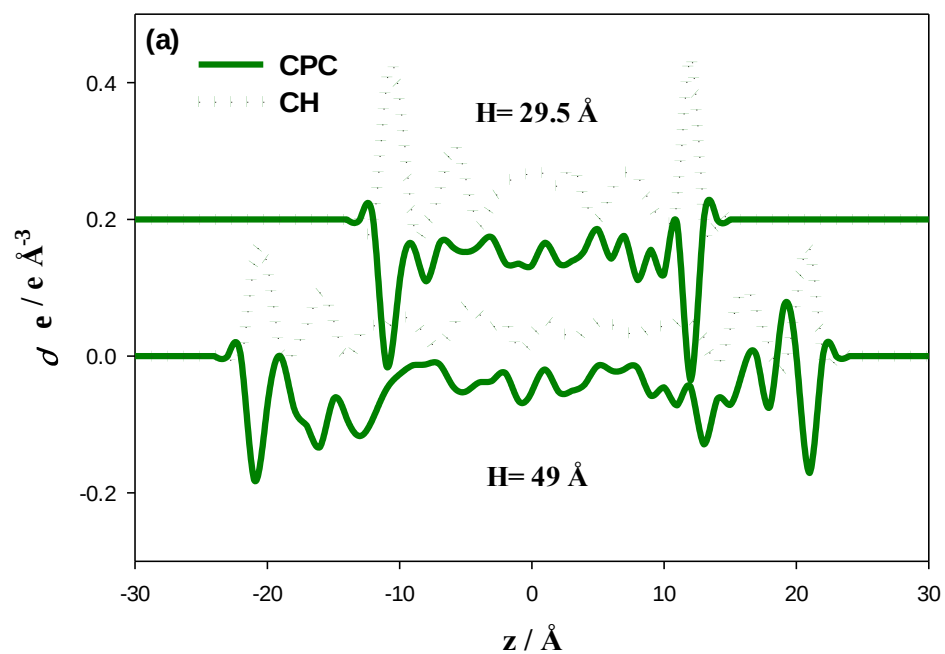


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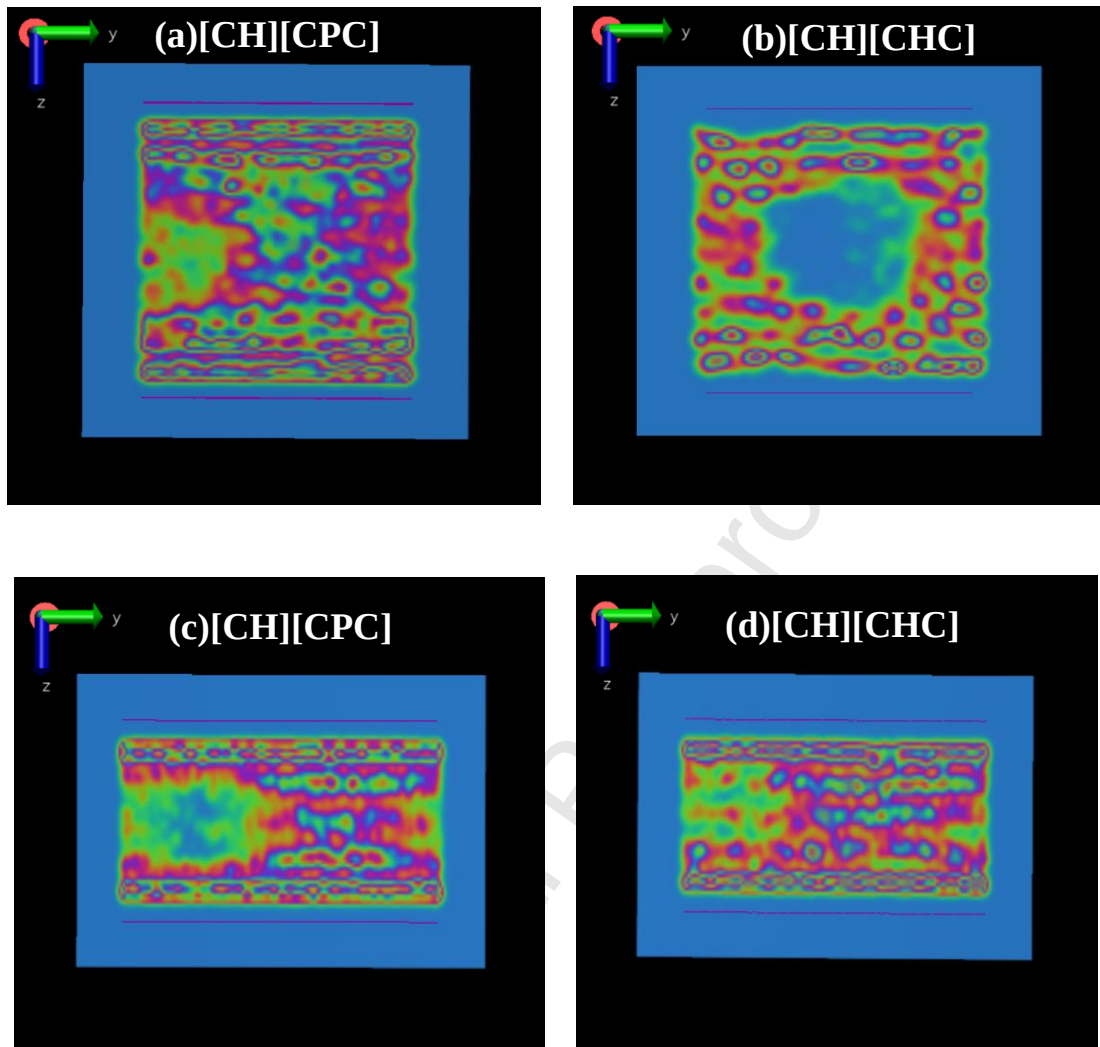


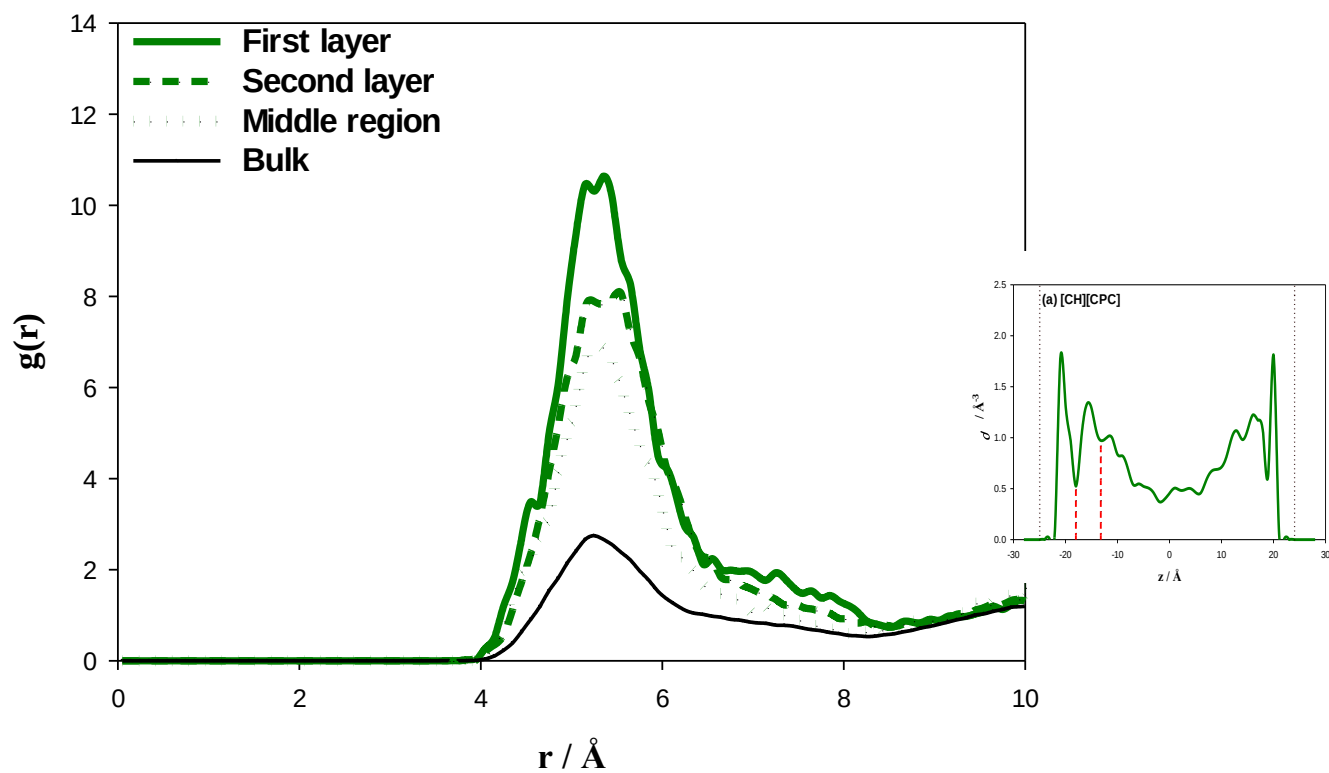
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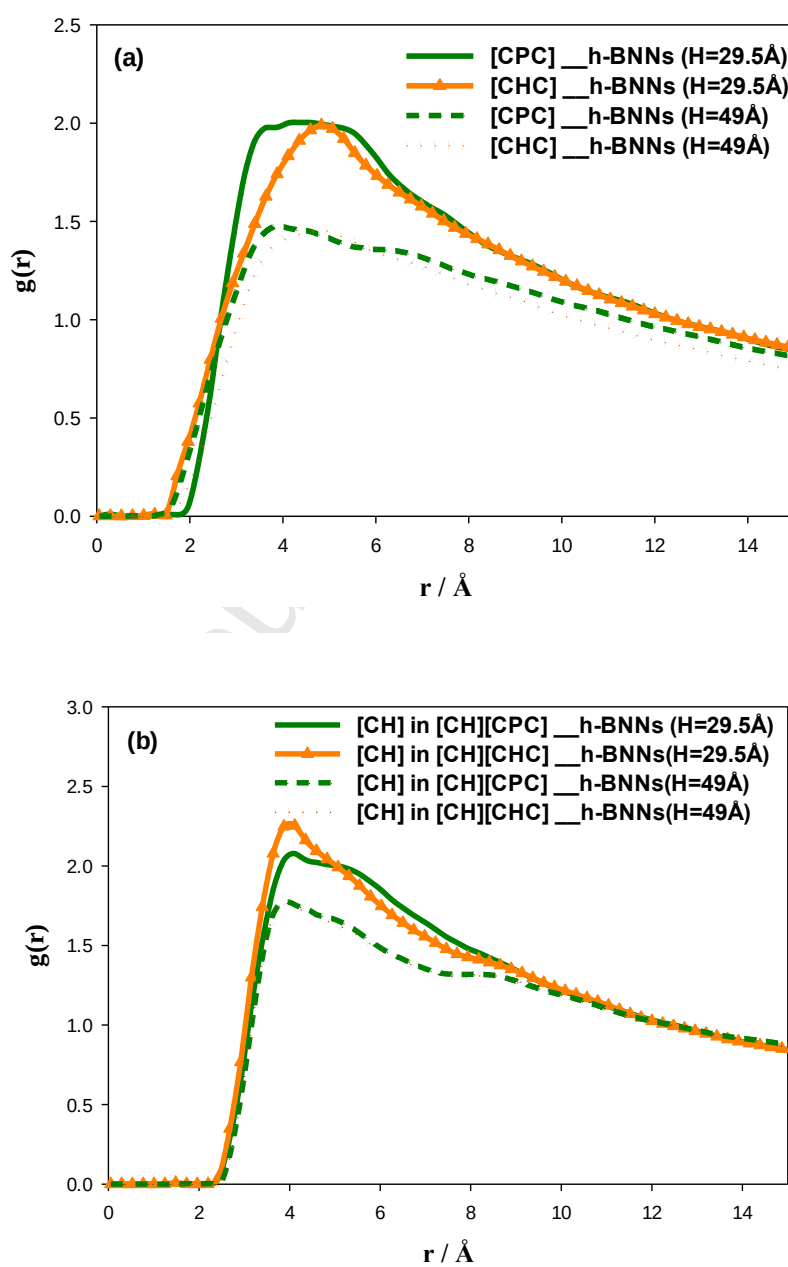


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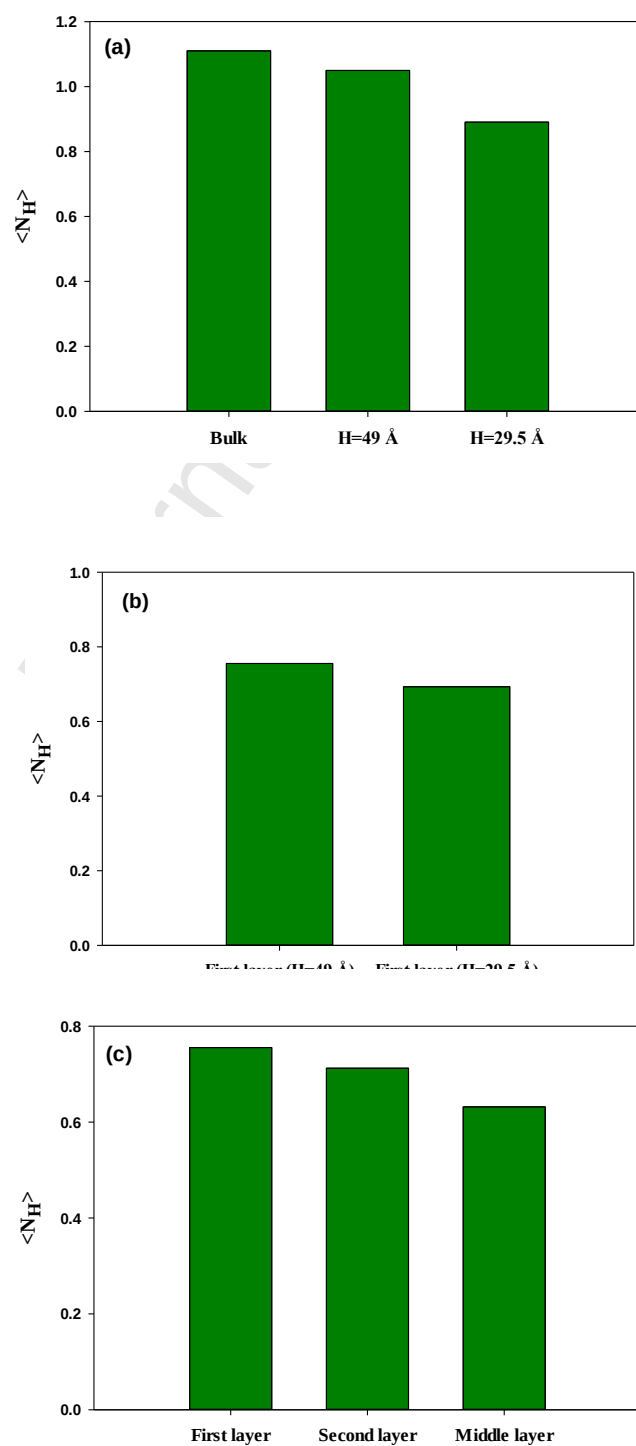


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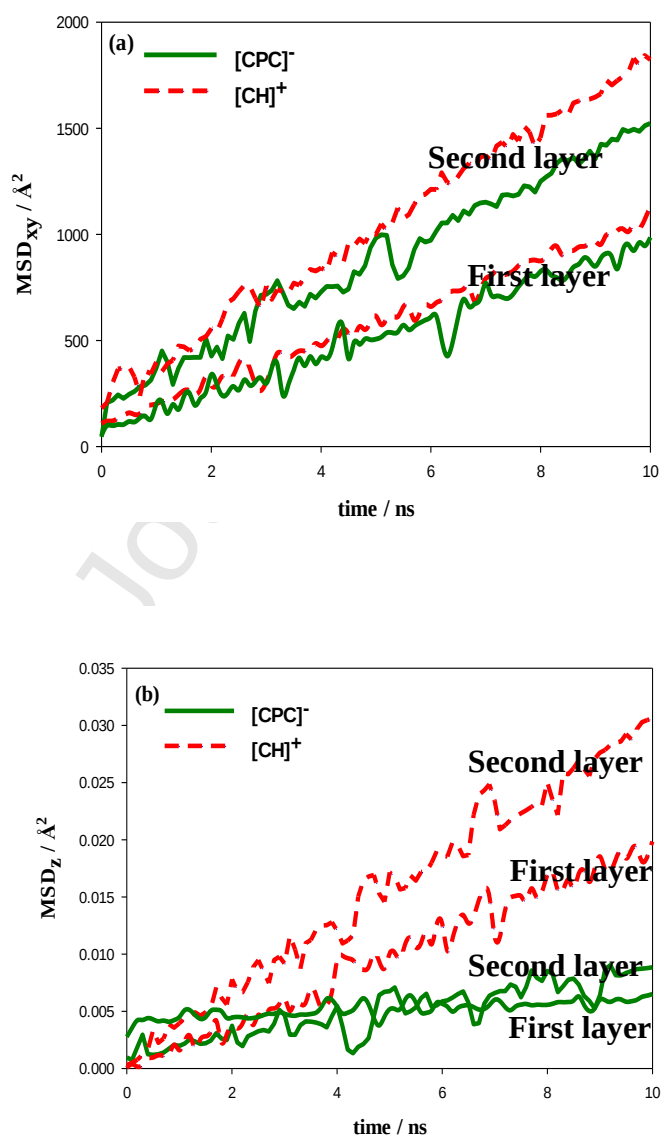
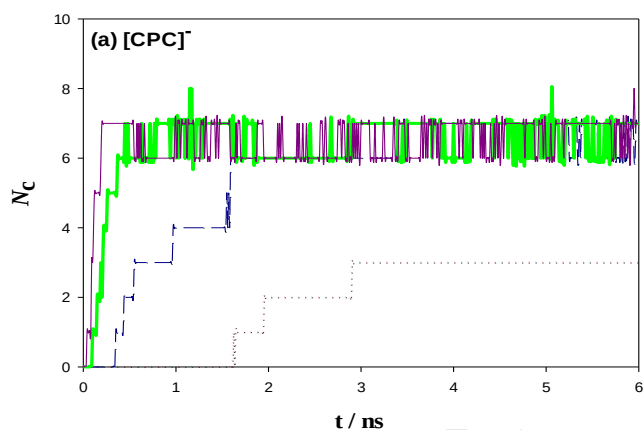
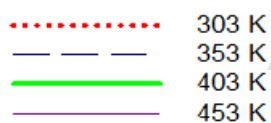


Fig. 9.



BNNT (10, 10)

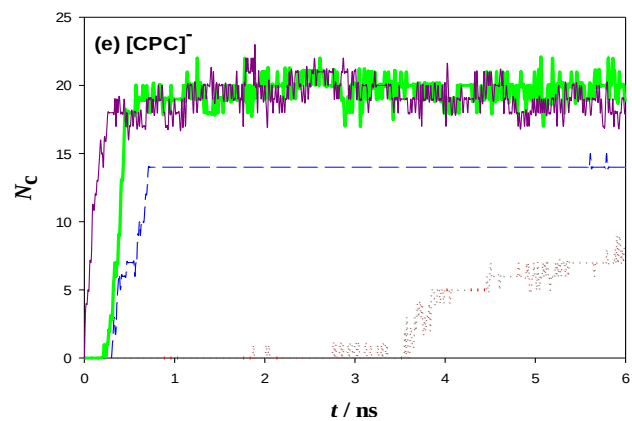
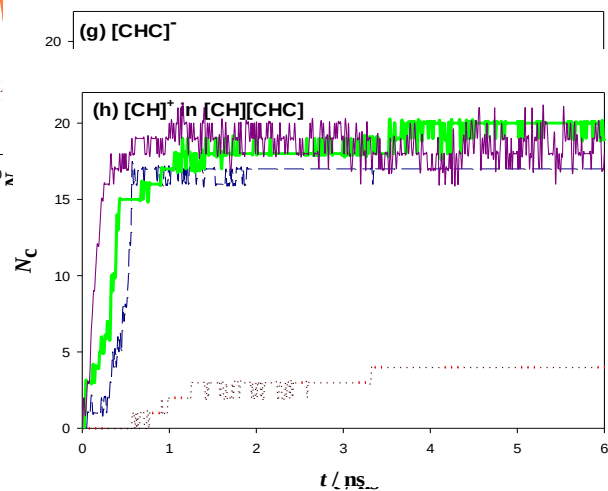
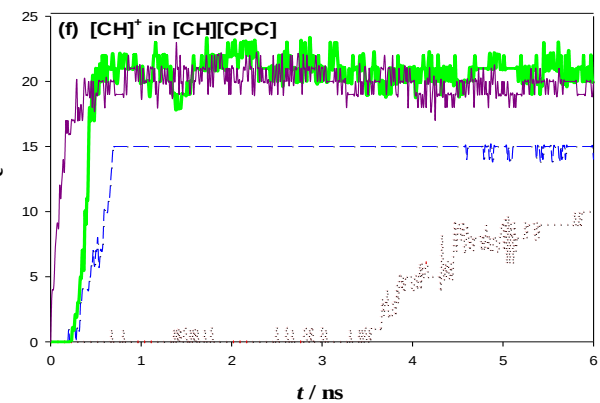
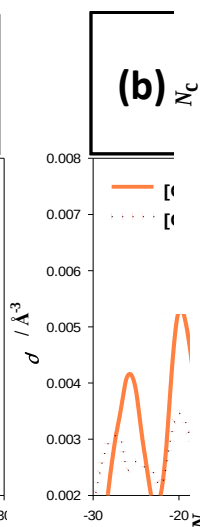
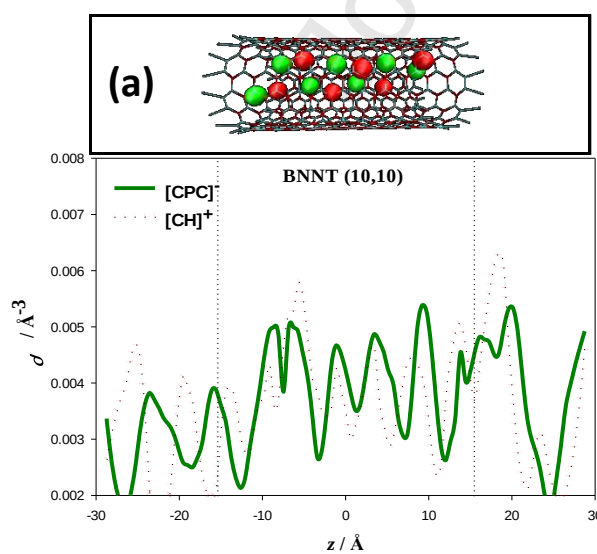


Fig. 10.



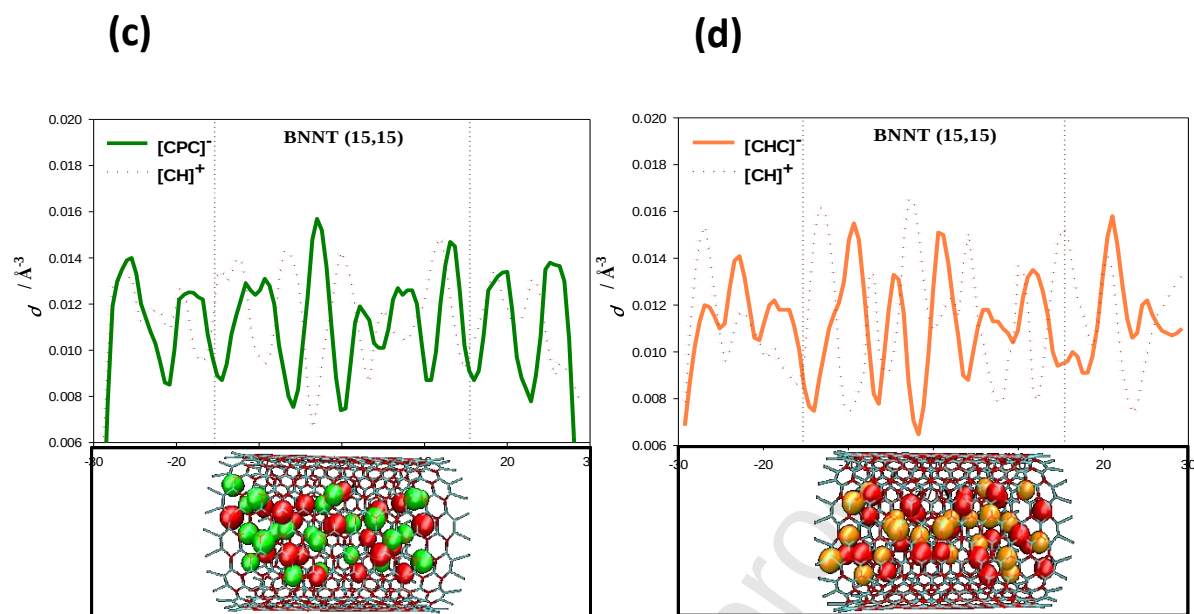


Fig. 11.

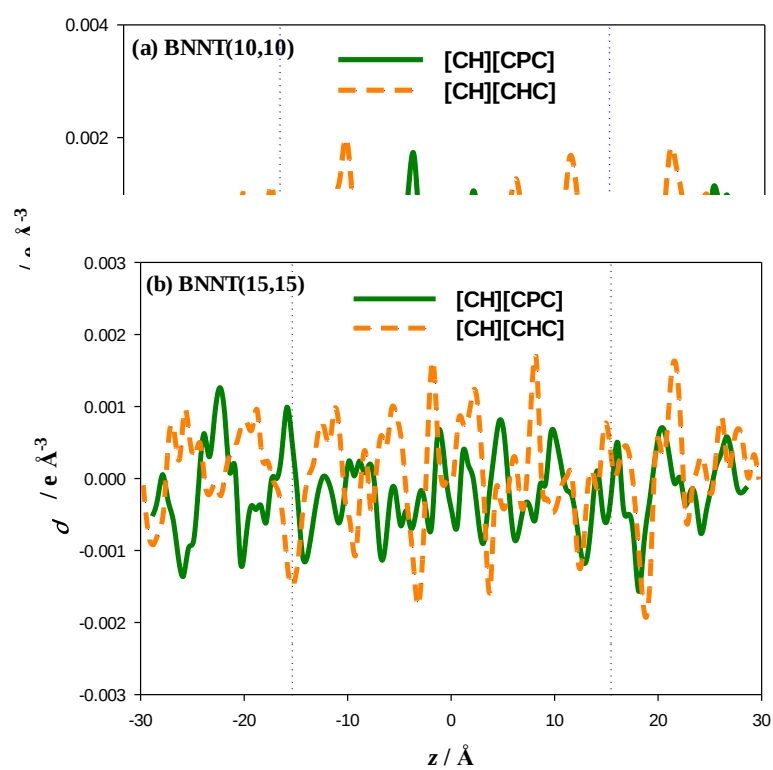
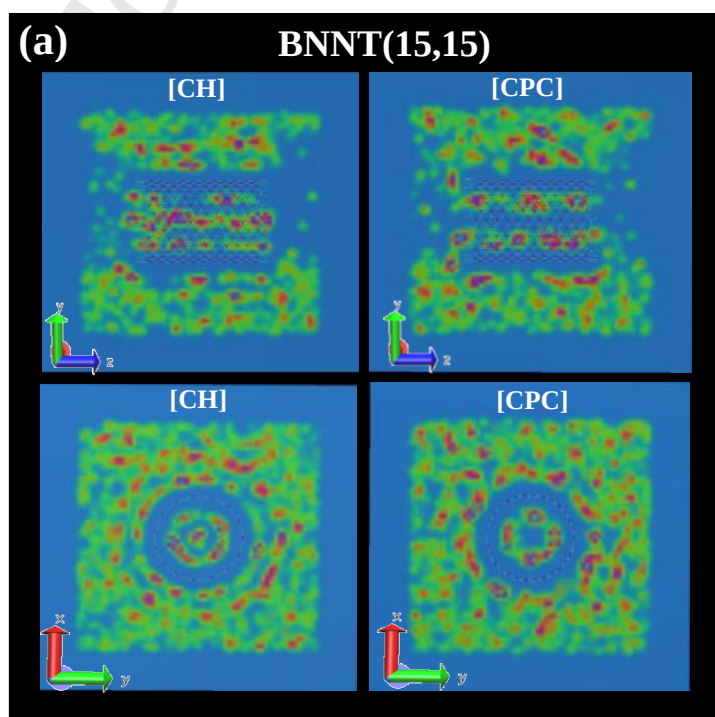


Fig. 12.



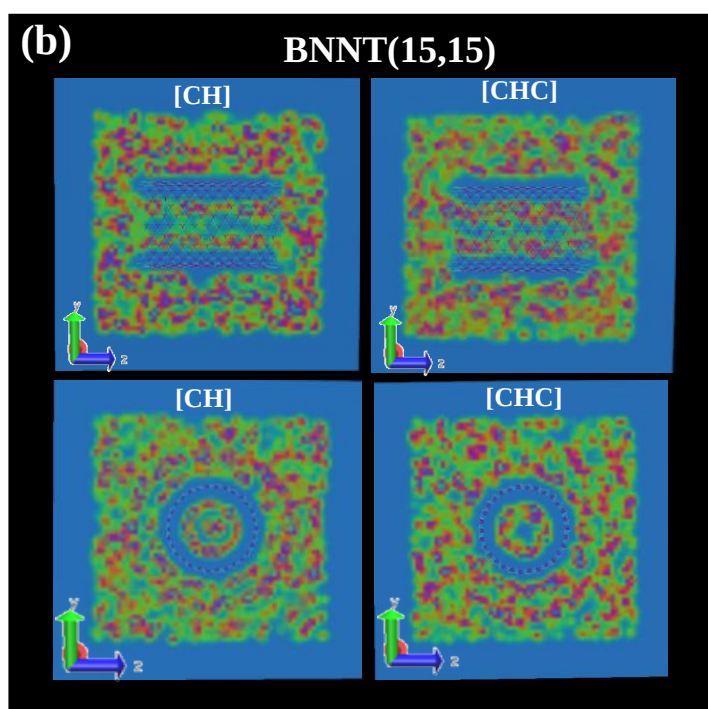
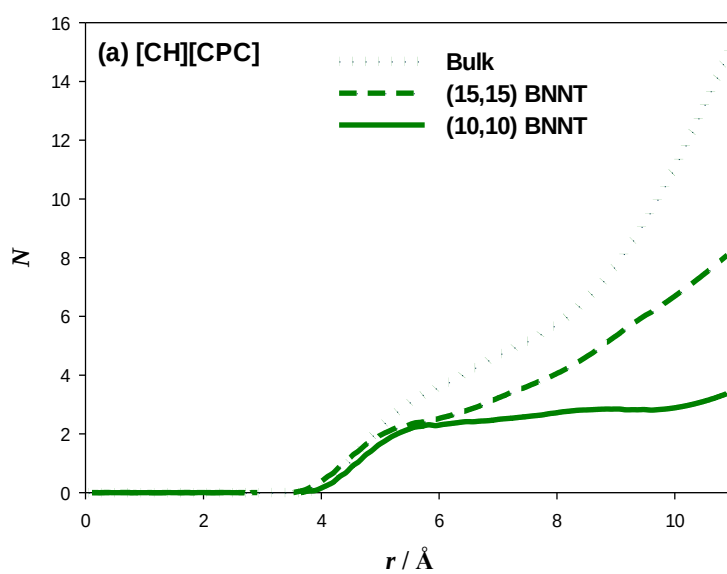


Fig. 13.



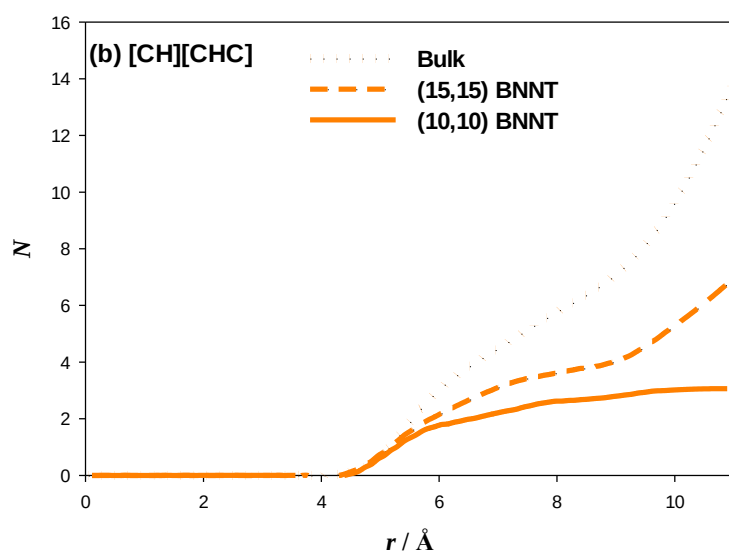
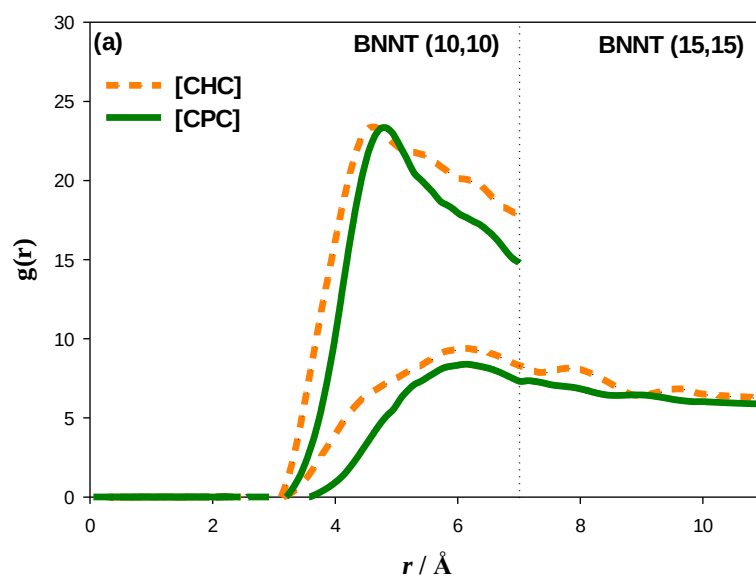


Fig. 14.



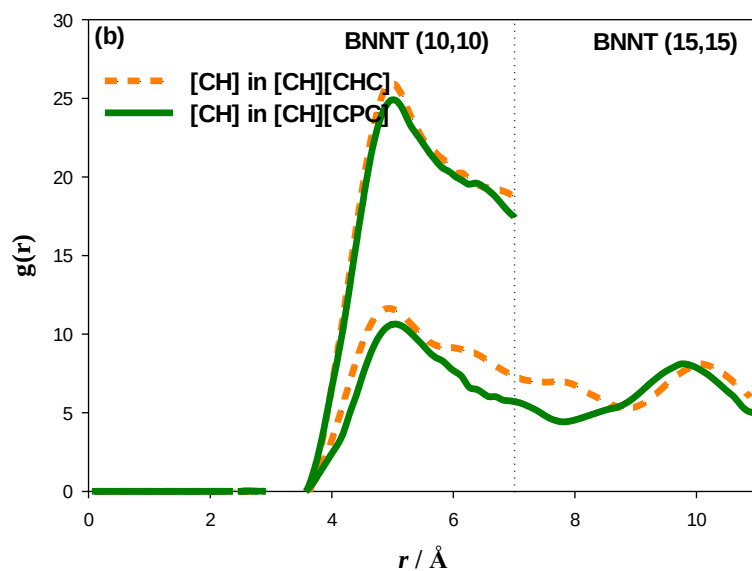


Table 1. Simulation details for the studied *h*-BNNSs

Interwall distances (Å)	Dimensions of each nanosheet (Å × Å)	Total of atoms of two nanosheets
29.5	71.38 × 63.62	1740
49	44 × 44	1584

Table 2. Simulation details for the studied BNNTs

Type of nanotube	BN atoms (H atoms)	Diameter ($D/\text{\AA}$)	Length ($L/\text{\AA}$)
BNNT (10,10)	480 (40)	13.6	30
BNNT (15,15)	720 (60)	20.3	30

Table 3. Number density values (ρ in nm^{-3}) of cations and anions inside the BNNT (n,n) at different temperatures and 1 bar

Ionic liquid	T (K)	BNNT (10,10)		BNNT (15,15)	
		ρ_{anion} (nm^{-3})	ρ_{cation} (nm^{-3})	ρ_{anion} (nm^{-3})	ρ_{cation} (nm^{-3})
[CH][CPC]	303	0.70	0.70	0.72	0.92
	353	1.40	1.40	1.44	1.54
	403	1.63	1.40	2.06	2.16
	453	1.63	1.63	1.96	2.06
[CH][CHC]	303	0.70	0.70	0.41	0.41
	353	1.40	1.40	1.65	1.75
	403	1.63	1.40	1.75	1.96
	453	1.63	1.40	1.65	1.85
[CH][SA]	303	0.00	0.23	0.10	0.10
	353	0.70	0.93	1.65	1.65
	403	1.63	1.63	2.06	1.96
	453	1.63	1.63	1.96	2.06
[CH][BE]	303	0.00	0.00	0.21	0.21
	353	0.93	1.16	1.65	1.65
	403	1.63	1.63	2.26	2.37
	453	1.86	1.63	2.37	2.26

Table 4. Self-diffusion coefficients along the axial direction, D_z , of the BNNTs for the center of mass of cations and anions in [CH][CPC] and [CH][CHC] inside the BNNT(n,n) and in the bulk ILs at 403 K and 1bar. β values are reported in parentheses for the time frame used in the calculations

n	[CH][CPC]		[CH][CHC]	
	D_z^+ ($10^{-11} \text{ m}^2 \cdot \text{s}^{-1}$)	D_z^- ($10^{-11} \text{ m}^2 \cdot \text{s}^{-1}$)	D_z^+ ($10^{-11} \text{ m}^2 \cdot \text{s}^{-1}$)	D_z^- ($10^{-11} \text{ m}^2 \cdot \text{s}^{-1}$)
10	1.11 (0.81)	0.32 (0.80)	0.71 (0.77)	0.25 (0.71)
15	1.94 (0.84)	0.70 (0.89)	0.83 (0.98)	0.33 (0.81)
bulk	6.31 (0.99)	3.72 (0.97)	4.80 (0.98)	3.59 (0.99)

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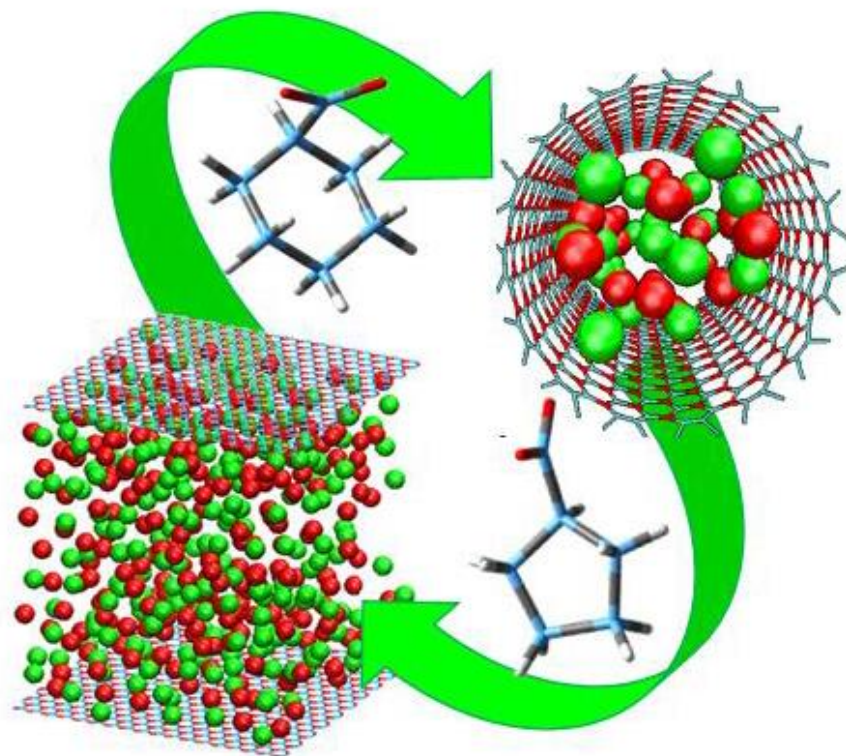
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Graphical abstract



Highlights:

- > Computational study on some ion gels (CBNAILs confined in the BN nanostructures) has been made.
- > The structure and dynamics of these ILs changes under confinement conditions.
- > The strength of ion–BN nanostructure interactions decreases with increasing nanopore size.
- > The dynamics of confined ions are slower than that of the bulk fluid and the ions move faster into the larger nanopores.
- > The results showed the ability of the CBNAILs to interact with BN nanosystems to develop the specific applications.